

Lecture 12

Lecturer: Asst. Prof. M. Mert Ankarali

12.1 The Kalman Decomposition

Intuition & Insight: The Kalman decomposition is a fundamental concept in linear systems theory that reveals the hidden internal structure of a system. By applying a specific similarity transformation, it separates the system dynamics into four distinct subsystems based on their reachability and observability properties. This allows us to structurally isolate the "minimal" part of the system that actually affects the input-output behavior.

In the reachability and observability lectures, we derived two types of standard forms, specifically for unreachable systems and unobservable systems (separately). Now, our goal is to propose a general standard form for an unreachable and unobservable system, based on the Kalman decomposition. The process is exactly the same for both DT and CT systems; thus, we will present the decomposition for only CT systems. Let

$$\dot{x} = Ax + Bu, \quad y = Cx + Du, \quad x \in \mathbb{R}^n$$

Let's assume that the system is neither reachable nor observable, and

$$\begin{aligned} \text{rank}[\mathbf{R}] &= r < n, \quad \text{range}[\mathbf{R}] = \mathcal{R} \\ \dim[\mathcal{N}(\mathbf{O})] &= \bar{o} > 0, \quad \bar{\mathcal{O}} = \mathcal{N}(\mathbf{O}) \end{aligned}$$

Let's consider the following similarity transformation:

$$\hat{A} = T^{-1}AT, \quad \hat{B} = T^{-1}B, \quad \hat{C} = CT, \quad \hat{D} = D$$

Let

$$T = [\begin{array}{c|c|c|c} T_{r\bar{o}} & T_{ro} & T_{\bar{r}\bar{o}} & T_{\bar{r}o} \end{array}]$$

Constructing the Transformation Matrix T : Let's define the sub-matrices as follows to systematically span the state space:

1. **Reachable and Unobservable ($T_{r\bar{o}}$):** Let $\mathcal{R}\bar{\mathcal{O}} = \mathcal{R} \cap \bar{\mathcal{O}}$, i.e., $x \in \mathcal{R}\bar{\mathcal{O}} \implies x \in \mathcal{R}$ and $x \in \bar{\mathcal{O}}$. Choose $T_{r\bar{o}}$ such that the columns of $T_{r\bar{o}}$ form a basis for $\mathcal{R}\bar{\mathcal{O}}$.
2. **Reachable and Observable (T_{ro}):** Choose T_{ro} such that $\text{Ra} [\begin{array}{c|c} T_{r\bar{o}} & T_{ro} \end{array}] = \mathcal{R} = \text{Ra}[\mathbf{R}]$, i.e., the columns of T_{ro} complement $T_{r\bar{o}}$ in the reachable subspace, and
3. **Unreachable and Unobservable ($T_{\bar{r}\bar{o}}$):** Choose $T_{\bar{r}\bar{o}}$ such that $\text{Ra} [\begin{array}{c|c} T_{r\bar{o}} & T_{\bar{r}\bar{o}} \end{array}] = \bar{\mathcal{O}} = \mathcal{N}[\mathbf{O}]$, i.e., the columns of $T_{\bar{r}\bar{o}}$ complement $T_{r\bar{o}}$ in the unobservable subspace.
4. **Unreachable and Observable ($T_{\bar{r}o}$):** Choose $T_{\bar{r}o}$ such that $\text{Ra}[T] = \mathbb{R}^n$. These columns complement the rest to span the entire n -dimensional state space.

Let's remember the important subspaces invariant under A and some critical features that will be helpful for constructing the Kalman decomposition:

$$\begin{aligned} A\mathcal{R} &\subset \mathcal{R} \\ A\bar{\mathcal{O}} &\subset \bar{\mathcal{O}} \implies A\mathcal{R}\bar{\mathcal{O}} \subset \mathcal{R}\bar{\mathcal{O}} \\ \text{Ra}[B] &\subset \mathcal{R} \\ \bar{\mathcal{O}} &\subset \mathcal{N}[C] \end{aligned}$$

Let's analyze the similarity transformation of the system matrix:

$$AT = T\hat{A}$$

$$A \left[\begin{array}{c|c|c|c} T_{r\bar{o}} & T_{ro} & T_{\bar{r}\bar{o}} & T_{\bar{r}o} \end{array} \right] = \left[\begin{array}{c|c|c|c} T_{r\bar{o}} & T_{ro} & T_{\bar{r}\bar{o}} & T_{\bar{r}o} \end{array} \right] \left[\begin{array}{c|c|c|c} A_{11} & A_{12} & A_{13} & A_{14} \\ \hline A_{21} & A_{22} & A_{23} & A_{24} \\ \hline A_{31} & A_{32} & A_{33} & A_{34} \\ \hline A_{41} & A_{42} & A_{43} & A_{44} \end{array} \right]$$

Let's expand $AT_{r\bar{o}}$:

$$AT_{r\bar{o}} = \left[\begin{array}{c|c|c|c} T_{r\bar{o}} & T_{ro} & T_{\bar{r}\bar{o}} & T_{\bar{r}o} \end{array} \right] \left[\begin{array}{c} A_{11} \\ \hline A_{21} \\ \hline A_{31} \\ \hline A_{41} \end{array} \right]$$

Since $\text{Ra}[T_{r\bar{o}}]$ is invariant under A (i.e., $A\mathcal{R}\bar{\mathcal{O}} \subset \mathcal{R}\bar{\mathcal{O}}$), $A_{i1} = 0$ for all $i > 1$.

Now let's focus on $A \left[\begin{array}{c|c} T_{r\bar{o}} & T_{ro} \end{array} \right]$:

$$A \left[\begin{array}{c|c} T_{r\bar{o}} & T_{ro} \end{array} \right] = \left[\begin{array}{c|c|c|c} T_{r\bar{o}} & T_{ro} & T_{\bar{r}\bar{o}} & T_{\bar{r}o} \end{array} \right] \left[\begin{array}{c|c} A_{11} & A_{12} \\ \hline 0 & A_{22} \\ \hline 0 & A_{32} \\ \hline 0 & A_{42} \end{array} \right]$$

Since $\text{Ra} \left[\begin{array}{c|c} T_{r\bar{o}} & T_{ro} \end{array} \right]$ is invariant under A (i.e., $A\mathcal{R} \subset \mathcal{R}$), $A_{32} = 0$ and $A_{42} = 0$.

Now let's focus on $A \left[\begin{array}{c|c} T_{r\bar{o}} & T_{\bar{r}\bar{o}} \end{array} \right]$:

$$A \left[\begin{array}{c|c} T_{r\bar{o}} & T_{\bar{r}\bar{o}} \end{array} \right] = \left[\begin{array}{c|c|c|c} T_{r\bar{o}} & T_{ro} & T_{\bar{r}\bar{o}} & T_{\bar{r}o} \end{array} \right] \left[\begin{array}{c|c} A_{11} & A_{13} \\ \hline 0 & A_{23} \\ \hline 0 & A_{33} \\ \hline 0 & A_{43} \end{array} \right]$$

Since $\text{Ra} \left[\begin{array}{c|c} T_{r\bar{o}} & T_{\bar{r}\bar{o}} \end{array} \right]$ is invariant under A (i.e., $A\bar{\mathcal{O}} \subset \bar{\mathcal{O}}$), $A_{23} = 0$ and $A_{43} = 0$.

As a result, $\hat{A}$ takes the form:

$$\hat{A} = \left[\begin{array}{c|c|c|c} A_{11} & A_{12} & A_{13} & A_{14} \\ \hline 0 & A_{22} & 0 & A_{24} \\ \hline 0 & 0 & A_{33} & A_{34} \\ \hline 0 & 0 & 0 & A_{44} \end{array} \right] = \left[\begin{array}{c|c|c|c} A_{r\bar{o}} & A_{12} & A_{13} & A_{14} \\ \hline 0 & A_{ro} & 0 & A_{24} \\ \hline 0 & 0 & A_{\bar{r}\bar{o}} & A_{34} \\ \hline 0 & 0 & 0 & A_{\bar{r}o} \end{array} \right]$$

Now let's focus on the input matrix transformation:

$$B = T\hat{B} = \left[\begin{array}{c|c|c|c} T_{r\bar{o}} & T_{ro} & T_{\bar{r}\bar{o}} & T_{\bar{r}o} \end{array} \right] \left[\begin{array}{c} B_1 \\ B_2 \\ B_3 \\ B_4 \end{array} \right]$$

Since $\text{Ra}(B) \subset \mathcal{R} = \text{Ra} \begin{bmatrix} T_{r\bar{o}} & T_{ro} \end{bmatrix}$, $B_3 = 0$ and $B_4 = 0$, thus $\hat{B}$ takes the form:

$$\hat{B} = \begin{bmatrix} B_{r\bar{o}} \\ B_{ro} \\ 0 \\ 0 \end{bmatrix}$$

Now let's focus on the output matrix transformation:

$$\begin{aligned} CT &= \hat{C} \\ C \begin{bmatrix} T_{r\bar{o}} & T_{ro} & T_{\bar{r}\bar{o}} & T_{\bar{r}o} \end{bmatrix} &= \begin{bmatrix} C_1 & C_2 & C_3 & C_4 \end{bmatrix} \\ \begin{bmatrix} CT_{r\bar{o}} & CT_{ro} & CT_{\bar{r}\bar{o}} & CT_{\bar{r}o} \end{bmatrix} &= \begin{bmatrix} C_1 & C_2 & C_3 & C_4 \end{bmatrix} \\ \begin{bmatrix} 0 & CT_{ro} & 0 & CT_{\bar{r}o} \end{bmatrix} &= \begin{bmatrix} C_1 & C_2 & C_3 & C_4 \end{bmatrix} \end{aligned}$$

and thus $\hat{C}$ takes the form:

$$\hat{C} = \begin{bmatrix} 0 & C_{ro} & 0 & C_{\bar{r}o} \end{bmatrix}$$

Key Insight - Input/Output Behavior: If we compute the transfer function of the transformed system, $\hat{H}(s) = \hat{C}(sI - \hat{A})^{-1}\hat{B} + \hat{D}$, we will find that it algebraically reduces to $H(s) = C_{ro}(sI - A_{ro})^{-1}B_{ro} + D$. This mathematically proves that the external input-output behavior is exclusively determined by the reachable and observable subsystem. The remaining dynamics are hidden from the input, the output, or both.

Based on the Kalman decomposition, we can obtain the reachable and observable (minimal) subsystem, the reachable-only subsystem, and the observable-only subsystem.

- **Reachable and observable subsystem**

$$\begin{aligned} \dot{x}_{ro} &= A_{ro}x_{ro} + B_{ro}u \\ y &= C_{ro}x_{ro} + Du \end{aligned}$$

- **Reachable-only (but not fully observable) subsystem**

$$\begin{aligned} \frac{d}{dt} \begin{bmatrix} x_{r\bar{o}} \\ x_{ro} \end{bmatrix} &= \begin{bmatrix} A_{r\bar{o}} & A_{12} \\ 0 & A_{ro} \end{bmatrix} \begin{bmatrix} x_{r\bar{o}} \\ x_{ro} \end{bmatrix} + \begin{bmatrix} B_{r\bar{o}} \\ B_{ro} \end{bmatrix} u \\ y &= \begin{bmatrix} 0 & C_{ro} \end{bmatrix} \begin{bmatrix} x_{r\bar{o}} \\ x_{ro} \end{bmatrix} + Du \end{aligned}$$

- **Observable-only (but not fully reachable) subsystem**

$$\begin{aligned} \frac{d}{dt} \begin{bmatrix} x_{ro} \\ x_{\bar{r}o} \end{bmatrix} &= \begin{bmatrix} A_{ro} & A_{24} \\ 0 & A_{\bar{r}o} \end{bmatrix} \begin{bmatrix} x_{ro} \\ x_{\bar{r}o} \end{bmatrix} + \begin{bmatrix} B_{ro} \\ 0 \end{bmatrix} u \\ y &= \begin{bmatrix} C_{ro} & C_{\bar{r}o} \end{bmatrix} \begin{bmatrix} x_{ro} \\ x_{\bar{r}o} \end{bmatrix} + Du \end{aligned}$$

Theorem: A state-space representation is minimal if and only if it is both fully reachable and observable.

Corollary: All minimal representations are algebraically equivalent.

Ex 12.1 Discrete-Time Example: Let a system be defined by:

$$x[k+1] = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 2 & 1 & 0 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & -1 & 0 & -1 \end{bmatrix} x[k] + \begin{bmatrix} -1 \\ 1 \\ 1 \\ 1 \end{bmatrix} u[k]$$

$$y[k] = \begin{bmatrix} 0 & 1 & 1 & 2 \end{bmatrix} x[k]$$

Obtain the Kalman decomposition and extract the reachable and observable (minimal) subsystem, the reachable-only subsystem, the observable-only subsystem, and the non-reachable and non-observable subsystem.

Step 1: Find the reachability matrix and reachable subspace

$$\mathbf{R} = \begin{bmatrix} A^3B & A^2B & AB & B \end{bmatrix} = \begin{bmatrix} 0 & 0 & 2 & -1 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & -2 & 1 \end{bmatrix}$$

$$\mathcal{R} = \text{Ra}[\mathbf{R}] = \text{Span} \left\{ \begin{bmatrix} -1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix} \right\}$$

Step 2: Find the observability matrix and non-observable subspace

$$\mathbf{O} = \begin{bmatrix} C \\ CA \\ CA^2 \\ CA^3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 & 2 \\ 2 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\bar{\mathcal{O}} = \mathcal{N}[\mathbf{O}] = \text{Span} \left\{ \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 0 \\ -1 \end{bmatrix} \right\} = \text{Span} \left\{ \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix} \right\}$$

Step 3: Find the intersection of the reachable and unobservable subspaces

One can see the intersection of the reachable and unobservable subspaces is one-dimensional, and

$$\mathcal{R}\bar{\mathcal{O}} = \text{Span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix} \right\}$$

Step 4: Construct the transformation matrix T and its inverse

Based on these findings, we can construct the transformation matrix as:

$$T = \begin{bmatrix} T_{r\bar{o}} & T_{ro} & T_{\bar{r}\bar{o}} & T_{\bar{r}o} \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & -1 & 0 \\ -1 & 1 & 0 & 1 \end{bmatrix}$$

$$T^{-1} = \begin{bmatrix} 1/2 & 1/4 & 1/4 & 0 \\ -1/2 & 1/4 & 1/4 & 0 \\ 0 & 1/2 & -1/2 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

Step 5: Compute the transformed state-space matrices

Now let's find the transformed state-space matrices and find the required decomposition:

$$\hat{A} = T^{-1}AT = \left[\begin{array}{c|c|c|c} A_{r\bar{o}} & A_{12} & A_{13} & A_{14} \\ \hline 0 & A_{ro} & 0 & A_{24} \\ \hline 0 & 0 & A_{\bar{r}\bar{o}} & A_{34} \\ \hline 0 & 0 & 0 & A_{\bar{r}o} \end{array} \right] = \left[\begin{array}{c|c|c|c} 0 & 2 & 1 & 3/2 \\ \hline 0 & 0 & 0 & 1/2 \\ \hline 0 & 0 & 0 & 1 \\ \hline 0 & 0 & 0 & 0 \end{array} \right],$$

$$\hat{B} = T^{-1}B = \left[\begin{array}{c} B_{r\bar{o}} \\ B_{ro} \\ 0 \\ 0 \end{array} \right] = \left[\begin{array}{c} 0 \\ 1 \\ 0 \\ 0 \end{array} \right]$$

$$\hat{C} = CT = [0 \mid C_{ro} \mid 0 \mid C_{\bar{r}o}] = [0 \mid 4 \mid 0 \mid 2]$$

Step 6: Extract the subsystems

We can then extract the reachable and observable (minimal), reachable-only, observable-only, and non-reachable and non-observable subsystems as:

- **Reachable and observable subsystem**

$$\begin{aligned} x_{ro}[k+1] &= [0]x_{ro}[k] + [1]u[k] \\ y[k] &= [4]x_{ro}[k] \end{aligned}$$

- **Reachable-only subsystem**

$$\begin{aligned} \begin{bmatrix} x_{r\bar{o}}[k+1] \\ x_{ro}[k+1] \end{bmatrix} &= \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_{r\bar{o}}[k] \\ x_{ro}[k] \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u[k] \\ y[k] &= [0 \mid 4] \begin{bmatrix} x_{r\bar{o}}[k] \\ x_{ro}[k] \end{bmatrix} \end{aligned}$$

- **Observable-only (but not fully reachable) subsystem**

$$\begin{aligned} \begin{bmatrix} x_{ro}[k+1] \\ x_{\bar{r}o}[k+1] \end{bmatrix} &= \begin{bmatrix} 0 & 1/2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_{ro}[k] \\ x_{\bar{r}o}[k] \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u[k] \\ y[k] &= [4 \mid 2] \begin{bmatrix} x_{ro}[k] \\ x_{\bar{r}o}[k] \end{bmatrix} \end{aligned}$$

- **Non-observable and non-reachable subsystem**

$$\begin{aligned} x_{\bar{r}\bar{o}}[k+1] &= [0]x_{\bar{r}\bar{o}}[k] + [0]u[k] \\ y[k] &= [0]x_{\bar{r}\bar{o}}[k] \end{aligned}$$